

Web Programming

Project Proposal

Interactive Pet Shop Management System

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Pet Shop Web Application - Project Proposal

1. Title

Interactive Pet Shop Management System

2. Scope

2.1. Project Overview

Development of a full-stack e-commerce web application for an online pet shop that allows users to browse pet categories, view pet details, add pets to cart, and complete purchases. The system will include an admin panel for inventory management and order tracking.

2.2. Key Features

- **Responsive Dashboard/Home Page:** Modern landing interface with navigation to different pet categories
- **Pet Categories:**
 - Wild Pets
 - Inside Home Pets
 - Fish
 - Birds
 - Domestic Pets
- **Interactive Sound Feature:** Click-to-play animal sounds for each pet
- **Database Integration:** Backend system for managing pet inventory, details, and user interactions

2.3. Technology Stack

- **Frontend:** HTML5, CSS3, JavaScript, React.js (optional)
- **Audio:** Web Audio API or Howler.js
- **Backend:** Node.js/Express or PHP
- **Database:** MySQL or MongoDB
- **Additional Libraries:** Bootstrap/Tailwind CSS for styling

2.4. Deliverables

- Fully functional web application with frontend and backend integration
 - Database schema and populated database
 - Project documentation
 - Source code with comments
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3. Work Breakdown Structure

Phase 1: Frontend Development

- **Dashboard & Home Page Design**
 - Create responsive layout
 - Design navigation system
 - Implement hero section
- **Shopping Cart Page**
 - Design cart interface showing all added items
 - Implement quantity controls (increase/decrease/remove)
 - Display price calculations including subtotal, tax, and total
 - Add 'Continue Shopping' and 'Proceed to Checkout' buttons
- **Checkout Page**
 - Design checkout form collecting customer details (name, email, phone, delivery address)
 - Create order summary section displaying all cart items and final total
 - Implement comprehensive form validation for all input fields
 - Add order confirmation message with order details
- **Pet Category Pages (Wild Pets, Fish)**
 - Design category layouts
 - Create pet cards/displays
 - Implement filtering options
- **Pet Category Pages (Birds, Inside Home Pets, Domestic Pets)**
 - Design category layouts
 - Create pet cards/displays
 - Implement filtering options
- **Interactive Audio Feature**
 - Implement audio player functionality
 - Add click-to-play sound for each pet
 - Optimize audio loading
- **UI/UX Enhancement**
 - Ensure responsive design across devices
 - Add animations and transitions
 - Polish overall user interface
- CSS framework setup and styling
- Testing and debugging
- Code review and integration

Phase 2: Database Integration

- **Database Design**
 - Create ER diagram
 - Design database schema
 - Set up database tables (Pets, Categories, etc.)
 - **Backend API Development**
 - Create RESTful API endpoints
 - Implement GET requests for pet data
 - Set up server configuration
 - **Pet Data Integration**
 - Connect frontend category pages to database
 - Display dynamic pet information
 - Implement search functionality
 - **User Interaction Features**
 - Implement "Add to Favorites" functionality
 - Create inquiry/contact form
 - Store user preferences in database
 - **Admin Panel (Optional Enhancement)**
 - Create basic admin interface
 - Implement CRUD operations for pets
 - Manage pet inventory
 - **Testing & Deployment**
 - Test database connections
 - Debug backend issues
 - Prepare for deployment
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- Database population with pet data
 - Frontend-Backend integration testing
 - Documentation and final presentation preparation
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4. Project Timeline

Phase	Duration	Milestones
Phase 1	Week 1-3	Complete frontend with static data
Phase 2	Week 4-6	Database integration and testing
Final Review	Week 7	Documentation and presentation

5. Success Criteria

- Responsive design working on desktop and mobile
- All pet categories displaying correctly
- Interactive sound feature functional
- Database successfully storing and retrieving pet information
- Clean, maintainable code with proper documentation